

Research on the Bidirectional Embedding Mechanism of Grassroots Administrative District Governance and Community Life Circle Services

— A Case Study of Elderly Meal Assistance Services in District P of City S¹

Xiong Jing^{1,2} Ding Yingjie¹ Wang Zihan³ Zhou Weijun⁴

(1. School of International and Public Affairs, Shanghai Jiao Tong University, Shanghai, 200030; 2. China Institute of Urban Governance, Shanghai Jiao Tong University, Shanghai, 200030; 3. Shanghai Star-River Bilingual School, Shanghai, 201108; 4. School of Geographic Sciences, East China Normal University, Shanghai, 200241)

【Abstract】 Against the backdrop of the rapidly expanding elderly population, elderly meal assistance services—positioned as the foremost priority among the twin challenges of "eating and medication administration"—bear directly upon the fundamental quality of life of older adults. District P, home to the largest elderly population in City S, is characterized by a complex demographic profile among its senior residents and correspondingly diverse demands for meal assistance services. In recent years, District P has encouraged all sub-districts and towns to pursue context-appropriate measures and foster exploratory innovation, resulting in steady improvements in the supply capacity, service modalities, and governance frameworks of meal assistance services. Notwithstanding these advances, a mounting challenge has emerged in the course of implementation: the growing demand among older adults for access to meal assistance services across administrative district boundaries. This is particularly pronounced among elderly cohorts whose household registration is decoupled from their actual place of residence, who receive elderly care support in non-native localities, or who reside in inter-administrative border zones. Taking District P as an empirical case, this paper examines the current state and attendant challenges of its elderly meal assistance services. Drawing upon the theoretical perspectives of collaborative governance, cross-boundary governance, and grassroots administrative district–community studies, it constructs an analytical framework for the bidirectional embedding of administrative district governance and community life circles. The study contends that by strengthening top-level design, promoting multi-stakeholder co-governance, reinforcing technological enablement, and unblocking demand feedback channels, it is feasible to advance the realization of high-quality cross-administrative sharing of elderly meal assistance services and to enhance older adults' sense of gain and well-being.

【Keywords】 Administrative District Governance; Community Life Circle; Bidirectional Embedding; Elderly Meal Assistance

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1 Problem Statement

Amidst rapid economic and social development, the degree of population aging in China has been steadily intensifying. In 2021, the proportion of the population aged 65 and above in China reached 14.2%, marking the country's entry into a moderately aging society; it is projected that China will transition into a stage of severe aging around 2035, with the aging rate surpassing 30%, and that by approximately 2050, both the absolute size and relative proportion of the elderly population will attain their historical peak^[1]. It is therefore evident that China's elderly population will continue to expand, and population aging will remain a fundamental national condition for a protracted period to come. The Report to the 20th National Congress of the Communist Party of China articulates the requirement to "implement the national strategy for actively responding to population aging, develop elderly care programs and industries, optimize services for childless and infirm elderly persons, and promote universal access to basic elderly care services for all older adults." The Central Economic Work Conference in 2024, the *Opinions on Deepening the Reform and Development of Elderly Care Services*, the Government Work Report in 2025, and the 4th Plenary Session of the 20th Central Committee have successively laid out important strategic arrangements for the development of the silver economy. Accordingly, the question of how to promote more adequate, balanced, and high-quality elderly care services in tandem with high-quality development—so as to enable older adults to live with enhanced quality, dignity, and happiness—merits sustained scholarly attention.

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Older adults are constrained by mobility, with their walking tolerance concentrated within a window of ten to fifteen minutes—a reality that imposes new imperatives for the proximate provision of high-quality, accessible elderly care services. In recent years, Chinese cities including Shanghai, Beijing, Jinan, Nanchang, and Wuhan have successively unveiled initiatives for the construction of "15-minute life circles" and "15-minute convenient living circles." These efforts aim to upgrade the essential services and public activity facilities required for daily life within a 15-minute walking radius (or a 15-minute driving radius within a township), covering approximately 3 to 5 square kilometers, thereby fostering age-friendly cities and cultivating safe, comfortable, and congenial community life circles. Elderly meal assistance services, as the most pressing of the two cardinal challenges of "eating and medication administration," bear directly on the basic quality of life of older adults, serve as a pivotal entry point for the development of the silver economy, and constitute an urgent practical issue that must be addressed in order to strengthen the supply capacity of elderly care services and 15-minute life circles.

Notwithstanding these endeavors, a constellation of complex issues persists in the actual implementation of elderly meal assistance services. At the territorial level, the spatial scope of certain 15-minute life circles encompasses areas that straddle administrative district boundaries. For instance, DML Sub-district and SL Town in District P, City S, owing to historical legacies, have evolved a pattern of "one locality under two jurisdictions" and "mutual territorial envelopment," which renders the distribution of public service resources exceedingly intricate and poses formidable challenges to the resource allocation of elderly meal assistance services. At the social level, the phenomena of decoupling between household registration and actual residence, and of receiving elderly care support in non-native localities, are pervasive, rendering it difficult for segments of the elderly population to access doorstep elderly care services such as meal assistance. At the service supply end, as the focus of social governance and services shifts downward to the grassroots level, each sub-district and town functions as the entity specifically tasked with delivering the "last mile" of elderly care services; nevertheless, resources are unevenly distributed across localities, resulting in divergent standards of meal assistance services from one jurisdiction to another. Relevant policies suffer from inadequate in-depth coordination: preferential subsidy policies vary across localities, and unified coordination mechanisms are lacking, impeding the formation of effective policy synergies. Information barriers persist as well: there is a dearth of integrated information platforms and information-sharing mechanisms, and the problem of cross-boundary settlement is conspicuous. Fundamentally, these issues stem from the tension that has emerged between grassroots administrative districts—characterized by rigid boundaries and territorialized governance—and life circles—defined by flexible boundaries and open mobility. The question of how to advance the bidirectional embedding of administrative districts and life circles through cross-boundary collaborative governance and, at the institutional level, resolve the difficulty of cross-administrative sharing of elderly meal assistance services, is an issue that urgently demands in-depth investigation.

2 Literature Review and Analytical Framework

2.1 Literature Review

First, with respect to the state of research on elderly meal assistance services, developed nations have accumulated comparatively extensive experience in this domain. In terms of service value, Alison et al. analyzed the positive impacts of the "Meals on Wheels" program on social connectedness, health services, and medical expenditures among high-need older adults^[2]. Regarding service model innovation, the "Watami no Takushoku" initiative operated by the Watami Group, proceeding from the standpoint of elderly convenience, optimizes the "meal selection—ordering—delivery" workflow and precisely tailors meal delivery plans to the needs of older adults. Domestic scholarship, by contrast, has focused predominantly on service models, resource allocation, and policy optimization. Ma Dongmei et al. principally synthesized and systematized relevant research in China from the perspectives of supply subjects and service models of elderly meal assistance, identified deficiencies in the perspectives, content, and methodologies of extant research, and advanced suggestions for future research directions^[3]. With respect to practical models, Zhong Huilan, taking Shanghai as a case in point, explored the diversified supply model and underscored the importance of collaborative participation by the government, the market, and social organizations^[4].

Second, concerning research on elderly care services within community life circles, the concept of the "life circle" can be traced back to the "neighborhood unit" advanced by American scholar Clarence Perry in 1929, who posited that it could cultivate a shared sense of social responsibility among residents and engender superior urban and social environments^[5]. Canadian scholar Bigonnesse et al. proposed a multidimensional theoretical framework for life circle

elderly care services, emphasizing that life circles must integrate medical, social, and leisure functions in order to support aging in place among older adults^[6]. In the early 1990s, China introduced the "life circle" concept into urban planning and development, and the notion of the 15-minute life circle gradually coalesced. In 2023, 13 departments including the Ministry of Commerce issued the Three-Year Action Plan for Comprehensively Promoting the Construction of 15-Minute Convenient Living Circles in Cities (2023–2025), calling for context-adapted measures and tailored strategies for each circle, with a focus on "the elderly and children," and for developing 15-minute convenient living circles into important vehicles for safeguarding and improving people's livelihood and for restoring and expanding consumption. Jiang Manqi and Tian Weiteng, taking Beijing, Tianjin, and Shanghai as case studies, examined the functional allocation characteristics of 15-minute community life circles in metropolises and proposed that elderly care service facilities should be integrated into life circle planning^[7]. Xiang Xupeng and Zhu Guoyi (2024), drawing upon the experience of Pudong New Area, emphasized the deep integration of life circles and elderly care services.

Finally, cross-administrative public service collaboration is emerging as a focal point of scholarly inquiry. In terms of theoretical frameworks and research significance, Ansell and Gash conducted a systematic examination of the practice and theoretical architecture of collaborative governance^[8]. Tao Xidong contended that it is imperative to adopt a mode of thinking oriented toward cross-regional, cross-level, and cross-departmental governance in order to construct a social cross-boundary governance system^[9]. Cai Zhibing and Zhang Keyun investigated the practical logic and five principal models of the reform for the moderate separation of economic zones and administrative districts, and highlighted the significance of cross-regional collaborative governance^[10]. With respect to governance models, Provan and Kenis categorized the network governance structures of cross-boundary cooperation among local governments into three archetypes: shared governance, lead organization governance, and administrative network governance^[11]. As for key determinants, Jones examined policy coordination in cross-state public service cooperation in the United States, emphasizing the importance of establishing unified standards. Concerning influencing factors, Xiong Jing, taking the Yangtze River Delta as a case study, analyzed the governance pathways of grassroots administrative districts adjacent to provincial boundaries, underscoring the necessity of policy coordination and resource integration^[12]. Luo Zhe and Shan Xuepeng observed that sub-districts and towns, as grassroots administrative districts, possess fiscal autonomy that reinforces the logic of territorial governance, yet this also gives rise to the problem of cost sharing for cross-regional service provision^[13]. In terms of practical models, Wang Yu et al. studied intergovernmental cooperation strategies in regional public service provision and proposed an optimization path of selective cooperation^[14].

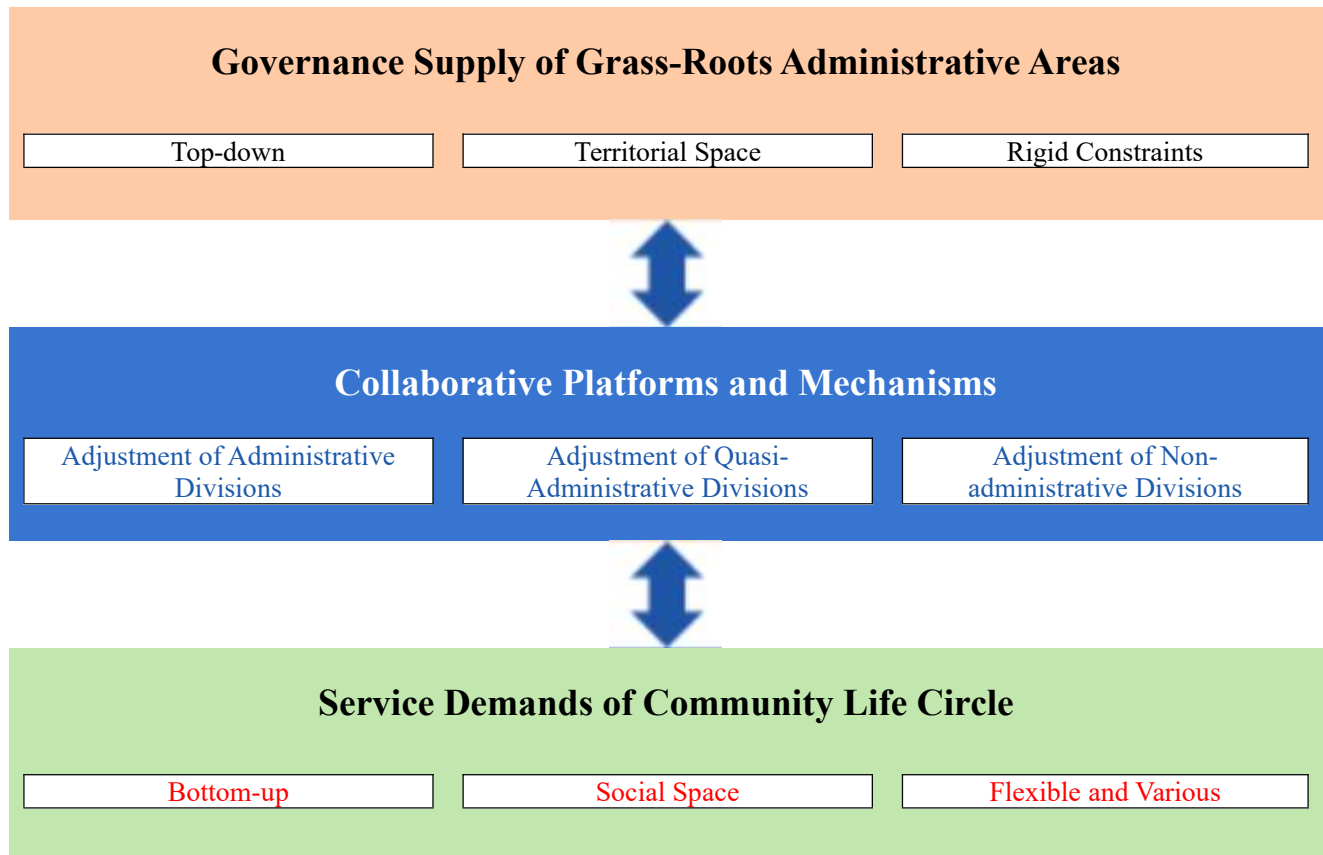
In sum, both domestic and international theoretical research and practical applications in these fields have yielded comparatively abundant results. Nevertheless, current scholarship still exhibits certain limitations, including constrained research perspectives and insufficient interdisciplinary engagement. This paper endeavors to integrate relevant theories, construct a bidirectional embedding analytical framework, and further pursue in-depth investigation and exploration into the problem of cross-administrative sharing of elderly meal assistance services.

2.2 Analytical Framework

Collaborative governance theory encompasses intergovernmental collaboration, regionalism, cross-sectoral partnerships, public service networks, consensus-building, and public participation^[15]. It emphasizes multi-stakeholder cooperation, resource integration, and institutional coordination, thereby offering theoretical guidance for addressing such challenges as the fragmentation of actors and the fragmentation of resources. Cross-boundary governance theory denotes a form of cross-sectoral cooperative governance that transcends administrative district boundaries and engages government departments, market forces, and the third sector^[16]. Focused on dismantling administrative regional barriers, it pursues the integration of cross-boundary public services through institutional innovation and factor mobility, directly responding to the policy incompatibilities and resource misallocations engendered by the fragmentation of administrative districts. Grassroots administrative district–community theory was advanced by Liu Junde in his exploration of administrative divisions and community systems; at its core is the optimization of the management systems of administrative districts and communities^[17], with a focus on the collaborative governance of the administrative authority of districts and the self-governance capacity of communities, and the construction of a sound interactive mechanism between government governance and community autonomy. Accordingly, anchored in collaborative governance as the core, cross-boundary governance as the foundation, and grassroots administrative

district–community theory as a point of reference, this paper develops an analytical framework comprising three dimensions—"administrative district governance, collaborative platform, and community life circle"—to address the practical predicament of cross-administrative sharing of elderly meal assistance services (as illustrated in Figure 1).

Figure 1 Analytical Framework for the Bidirectional Embedding of Administrative District Governance and Community Life Circles



Source: Compiled by the authors

First, the administrative district falls within the domain of political space; it constitutes a politico-geographic unit demarcated by the state for purposes of governance and is vested with statutory boundaries. Administrative districts discharge administrative management functions, and grassroots administrative districts strictly implement directives issued by superior authorities and remain accountable to higher-level organs. Administrative decentralization, however, may engender horizontal competition among local administrative districts and give rise to a pattern of "vigorous competition, lukewarm cooperation"^[13]; concomitantly, there exist objective micro-level self-interested orientations and inter-actor resource competition. Consequently, with respect to top-level design, it is imperative to break through the confines of administrative division boundaries through structural embedding—that is, through institutional design and policy integration on the part of the administrative system—establish a meal assistance service system attuned to population mobility, and proactively align with the practical needs of community life circles.

Second, the life circle pertains to the category of social space; it is a functionally radiating unit forged and propelled by the daily needs of residents, with comparatively fluid boundaries. The formation of a life circle proceeds in a bottom-up manner: within a defined scope, it furnishes the functions required by residents, such as daily life services, social and recreational activities, and consumer shopping. Life circles are not hierarchically subordinate to one another, and when the functional scope of a life circle extends beyond that of a single administrative district, it generates demands for cross-district collaboration. Functional embedding foregrounds the articulation of demands and the role of social capital within community living areas, which in turn exert a reciprocal influence on the policy adjustment and service innovation of administrative districts, thereby facilitating the upward transmission of grassroots dynamism.

Finally, the pivotal coupling point of the bidirectional embedding mechanism resides in the establishment of a collaborative platform. When the spatial configuration of residents' demands is misaligned with that of administrative

provision, adjustments must be effected through collaborative platforms or mechanisms. Such adjustments may be achieved through the restructuring of grassroots administrative districts, through quasi-administrative district adjustments, or through alternative administrative district governance mechanisms that do not entail zoning modifications. The overarching objective is to attain a match between the supply of grassroots administrative district governance and the demand for community life circle services through spatial integration, thereby mitigating spatial friction and enhancing the accessibility, equalization, and sustainability of public services. It bears noting that, against the backdrop of escalating costs associated with administrative division adjustments, non-zoning-adjustment approaches to grassroots administrative district governance—that is, the construction of cross-administrative collaborative mechanisms, and particularly those empowered by digital intelligence—afford superior operability and efficiency.

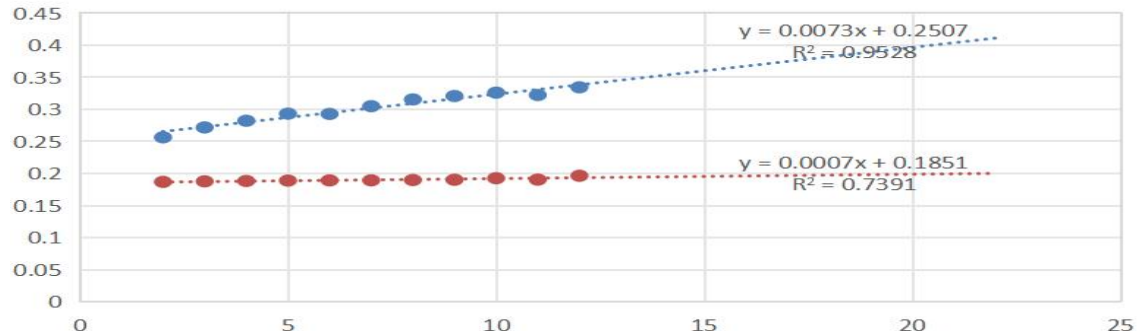
3 Case Analysis

3.1 Basic Overview of Elderly Meal Assistance Services in District P, City S

District P, as the administrative district with the largest elderly population in City S, encompasses a diverse array of area types, including premium urban districts, modern towns, and picturesque villages. The proportion of non-local permanent residents reaches 40%, and the phenomenon of elderly migrants without local household registration relocating for old-age support is pervasive; the elderly population exhibits a complex demographic structure, and the demand for elderly meal assistance services is highly diversified. These factors have exacerbated the service supply contradictions stemming from the "separation of household registration and place of residence" and "cross-regional elderly care," rendering the issue of cross-administrative sharing particularly emblematic. Although elderly meal assistance services in District P have been steadily enhanced in terms of supply capacity, service models, and governance systems, District P has a complex administrative division structure (comprising 12 sub-districts and 24 towns), with pronounced disparities in fiscal resources and social assets across sub-districts and towns, as well as divergent meal assistance subsidy policies and preferential treatment levels. Furthermore, fiscal accounts among sub-districts and towns are settled on a relatively independent basis, information silos remain to be dismantled, resources cannot be efficiently integrated, and the demand of older adults for cross-administrative sharing of meal assistance services remains unmet.

From the vantage point of the evolving characteristics of the elderly population in District P (as depicted in Figure 2), between 2013 and 2023, the proportion of the registered elderly population aged 60 and above in District P rose from 25.62% to 33.49%, marking an increase of nearly 390,000 persons and displaying a strongly correlated upward trajectory. Linear regression projections indicate that this proportion will reach 37.84% by 2030. As the degree of population aging intensifies, the problem of "meal access difficulties" among older adults has become increasingly salient. To address the demand for meal assistance—particularly for the advanced-age elderly, those living alone, empty-nest elderly persons, as well as all-elderly households and families who have lost their only child—City S has incorporated elderly meal assistance services as a vital component of its elderly care service practical projects, promulgated a series of policies, and progressively established a comparatively comprehensive elderly meal assistance service system. Under the auspices of the Leading Group for the Development of the Social Elderly Care Service System of District P, District P has set up a District Elderly Meal Assistance Service Working Group, whose membership comprises the District Civil Affairs Bureau (lead unit), the District Community Work Party Committee, the District Development and Reform Commission, the District Finance Bureau, the District Market Supervision Bureau, and all sub-districts and towns. At the policy level, the *Implementation Plan of District P on Promoting and Regulating Elderly Meal Assistance Services* was issued in 2021, followed by the *Service Specifications for Elderly Meal Assistance Points* in 2022. Additionally, the Fiscal Support Opinions for the Development of Elderly Care Services During the 14th Five-Year Plan Period in District P was promulgated in 2021, which, in adherence to the principles of "universal access, gradient design, and targeted subsidies," gives full rein to the government's guiding role and the principal role of social forces.

Figure 2 Trend of the Proportion of Elderly Population in District P



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Linear (Proportion of the Total Population in District P)	Linear (Proportion of the Total Population in the City)

Source: Compiled by the authors

The elderly meal assistance service system in District P has undergone continuous refinement. In terms of site expansion, as of the end of 2024, District P had established 489 elderly meal assistance service venues, comprising 55 community (senior) canteens, 395 elderly meal assistance points (including micro-meal assistance outlets), and 39 designated and accredited social catering enterprises, essentially achieving district-wide coverage. With respect to meal services, each sub-district and town has pursued diversified explorations tailored to local conditions in areas such as meal offerings, pricing, smart meal assistance, and the integration of multiple functions, thereby advancing the transformation of elderly dining from "having access to meals and being adequately fed" to "dining well and eating nutritiously and healthily." In terms of financial safeguards, pursuant to the *Fiscal Support Opinions for the Development of Elderly Care Services During the 14th Five-Year Plan Period in District P*, the district finance provides corresponding construction and operation subsidies to meal assistance entities that satisfy assessment criteria, with subsidy funds earmarked exclusively for their designated purposes; each sub-district and town formulates meal fee subsidy policies in accordance with actual circumstances. Financial coordination is strengthened through a model of multi-party support and diversified participation characterized by "individuals contribute a share, enterprises concede a share, the government subsidizes a share, and charitable organizations donate a share."

3.2 Challenges Confronting the Service System

3.2.1 Institutional Barriers

The rigid constraints imposed by the household registration system and fiscal appropriation mechanisms constitute the primary institutional barriers. On the one hand, the allocation of public service resources under the prevailing system is largely predicated on the registered population, and meal assistance subsidies in District P are frequently circumscribed to the locality of household registration or permanent residence. Older adults who are disinclined to forfeit their preferential entitlements opt to dine at meal assistance points in their immediate vicinity, thereby precluding older adults from other sub-districts and towns from gaining access to high-quality meal assistance venues distinguished by superior service standards, diverse culinary offerings, and age-friendly amenities. While the demand among some older adults for inter-sub-district and cross-venue dining remains unmet, the elderly meal assistance industry is unable to generate a "catfish effect," which militates against the enhancement of overall service quality and the innovative advancement of the sector. On the other hand, as the focus of social governance and service delivery shifts downward to the grassroots level, each sub-district and town concretely assumes responsibility for elderly care services. It is required to independently mobilize property resources, formulate preferential subsidy policies, and disburse subsidy funds in accordance with its own level of socioeconomic development, thereby engendering a path dependency characterized by "funds follow the household registration." Against the backdrop of fiscal decentralization, disposable revenue within a given jurisdiction has emerged as one of the principal resources for inter-administrative competition. Since funding for meal assistance services derives predominantly from local fiscal coffers, local governments risk falling into the trap of "cost-benefit calculus" in cross-administrative cooperation: they are neither willing to defray costs on behalf of other regions nor receptive to the prospect of local resource outflows eroding service efficiency.

3.2.2 Managerial Barriers

The challenges of mutual qualification recognition, standard harmonization, and regulatory coordination entailed by cross-regional service provision are particularly acute. Empirical investigations reveal that disparities obtain in the qualification criteria, service quality requirements, and price subsidy levels of meal assistance service institutions across different administrative districts. The absence of cross-regional mutual recognition mechanisms results in formidable impediments to service convergence. For instance, there are as many as 36 distinct sets of management standards governing meal assistance venues in District P, and consensus has yet to be forged, giving rise to the "five inconsistencies" predicament: inconsistent professional access standards, inconsistent operational standards, inconsistent supervisory standards, inconsistent reward and punishment management standards, and inconsistent rectification protocols. These discrepancies render the regulatory system difficult to operationalize and impede its

comprehensive implementation. Concomitantly, controversies have arisen concerning the fairness of competition: the relevant subsidy policies for elderly meal assistance have provoked discontent among certain conventional catering enterprises, which contend that the government is undermining fair market competition and inflicting adverse impacts on their businesses. Furthermore, owing to lacunae and vulnerabilities in routine supervision and management, some enterprises exploit regulatory loopholes, precipitating the phenomenon of "policy arbitrage." By way of illustration, during an operational assessment of senior canteens in District P conducted by the Minfu Elderly Care Service Center of City S in 2023, it was uncovered that at a community senior canteen in SY Town, the reported designed meal-serving capacity (500–799 meals per day) bore no reasonable correspondence to the combined floor area of the kitchen and storeroom; the on-site audit of average daily elderly patronage revealed that actual consumption amounted to a mere 54.9% of the average daily meal supply volume reported in 2022.

3.2.3 Resource Barriers

Uneven resource endowments across regions are prone to precipitating interest-driven contestation, and the resource constraints confronting economically disadvantaged areas are especially severe. In geographically remote and poorly connected localities, numerous older adults face formidable barriers to accessing meal assistance services. As an enduring public welfare undertaking, elderly meal assistance necessitates sustained financial support—a burden that some economically straitened villages and communities struggle to bear over the long term. For enterprises, the elderly meal assistance industry falls into the category of "low-profit, public-benefit services," with inherently circumscribed profit margins. Establishing chain outlets across regions escalates costs across multiple dimensions, including procurement and distribution, and additionally entails duplicative approval procedures at the sites of branch operations. Enterprises are consequently more disposed toward scaled-up service provision within their own jurisdictions than cross-regional expansion. Furthermore, smart service platforms across sub-districts and towns are mutually incompatible and disconnected, and unified technical standards and data interfaces are wanting, which constrains the efficient integration and optimal allocation of various factors and resources. This state of affairs not only precludes the verification of older adults' identity credentials and the confirmation of subsidy entitlements but also impedes the sharing of pertinent data such as dining records, dietary requirements, and health profiles of elderly patrons. Data interoperability has yet to be achieved across government departments including civil affairs, health, finance, and market supervision; data pertaining to the health status of older adults, subsidy policies, settlement records, and inspection outcomes are dispersed among disparate agencies, and fragmented governance has eventuated in the squandering of resources.

3.2.4 Cultural Barriers

Territorially bounded mindsets and disparities in administrative culture likewise constrain cross-regional cooperation. On the one hand, the governance of grassroots administrative districts is characterized by territorialization; to facilitate spatial planning and management with clearly delineated powers and responsibilities, meal assistance venues are sited and services delivered along administrative demarcations. This state of affairs imperceptibly circumscribes the temporal and spatial parameters of older adults' daily lives and tends to produce a consolidating effect of the "administrative district as life circle." The long-entrenched territorial management ethos of "each tending to one's own doorstep" resists rapid transformation, and divergences in policy proclivities, implementation benchmarks, and administrative conventions across different administrative districts further compound collaboration costs. Simultaneously, certain older adults evince low receptivity to novel service modalities; some elderly individuals who either lack smartphones or lack proficiency in their operation are unable to authenticate personal information, place orders and effect payments, or conduct information inquiries via smart platforms. A proportion of older adults harbor psychological reservations about practices such as identity information binding and facial recognition payment, and remain heavily reliant on traditional service modes including cash payment and telephone reservations, thereby constituting obstacles at the psychocultural level.

4 Countermeasures and Recommendations Grounded in the Bidirectional Embedding Mechanism of Administrative District Governance and Community Life Circles

4.1 Strengthening Top-Level Design and Consolidating Policy Safeguards

The restructuring of the policy architecture constitutes an indispensable safeguard for the efficacious operation of the bidirectional embedding mechanism, with its principal thrust lying in the dismantling of policy barriers and the refinement of institutional norms. First, it is imperative to articulate cross-administrative interest compensation mechanisms so as to mitigate regional fiscal inequities. To allay local government apprehensions that "resource outflows" will exacerbate fiscal burdens, explorations may be undertaken of such instruments as fiscal transfer

payments and pilot tax-sharing arrangements, thereby galvanizing local enthusiasm for participation in the cross-administrative provision of elderly meal assistance services. Second, relevant standards require harmonization to surmount the quandary of institutional fragmentation. It is recommended that provincial (municipal) authorities promulgate foundational specifications for elderly meal assistance services, encompassing both mandatory requisites—including industry access standards, operational standards, supervisory standards, and rectification protocols—so as to elevate the overall caliber of service management within the elderly meal assistance sector; and flexible benchmarks such as the price band for guaranteed meals, the subsidy range for basic services, the age stratification of subsidy recipients, and the modalities of subsidy disbursement, thereby attenuating disparities in basic meal assistance policy standards among local governments. Third, list-based management should be advanced, and entry and exit mechanisms refined. The establishment of a "whitelist" and "blacklist" system is proposed: social catering establishments that satisfy operational criteria and have maintained a clean complaint record for five consecutive years may be inscribed on the meal assistance service "whitelist" or even enlisted to participate in the operation of senior canteens, thereby incentivizing the proliferation of high-quality social dining venues; operating entities characterized by substandard food quality, or worse, food safety infractions, predatory pricing practices, and elevated complaint frequencies shall be placed on the "blacklist" and barred from entry into the elderly meal assistance service sector for a stipulated period or on a permanent basis.

4.2 Advancing Multi-Stakeholder Synergy and Cultivating a Paradigm of Co-Construction, Co-Governance, and Co-Benefit

Multi-stakeholder synergy serves as the social foundation for the effective functioning of the bidirectional embedding mechanism. Insofar as elderly meal assistance services implicate a multiplicity of actors, it is incumbent upon all parties to intensify communication and collaboration, clarify their respective responsibilities and functional divisions, and forge cross-administrative service synergies by constructing an elderly meal assistance service network predicated on the principle that "the government subsidizes a share, enterprises concede a share, society contributes a share, and individuals defray a share." The government bears overarching responsibility for elderly meal assistance services and should discharge its functions in policy guidance, organizational support, resource integration, platform development, and regulatory oversight; it should foster mutual trust among local governments and across functional departments, and expand the supply of universal, foundational, and safety-net elderly meal assistance services. Enterprises should fully leverage their role as market entities, deepen their engagement in niche segments of the elderly meal assistance industry, and progressively emerge as the principal force in service delivery, furnishing older adults with increasingly professional, personalized, and refined meal assistance services and products, satisfying the diversified and multi-tiered needs of the elderly population, unlocking the consumption potential of older adults, promoting the industrialization and brand development of elderly meal assistance services, and catalyzing the expansion of the silver economy. The role of social forces in public service delivery is assuming growing prominence, as they can redress gaps and unmet needs in the services provided by government and enterprises. Catering associations, experts, and scholars should be mobilized to furnish technical guidance at every juncture of service delivery, while active older adults and workers in new employment forms should be integrated into the volunteer workforce. Older adults and their family members, as both beneficiaries and consumers, should be reached through multifarious channels and modalities to disseminate and implement policies, ensuring that a greater number of older adults can genuinely access the services on offer. Concurrently, family account schemes should be explored, and family meal packages and weekend promotions introduced to stimulate intergenerational family consumption.

4.3 Technology-Empowered Smart Governance and Data-Driven Services

Technological enablement constitutes a critical pillar underpinning the effective operation of the bidirectional embedding mechanism. It achieves this primarily through digital modalities that reduce transaction costs associated with cross-regional, cross-level, cross-departmental, and multi-stakeholder collaboration, while concomitantly enhancing service precision and accessibility. First, a smart digital platform should be established. Provincial (municipal) authorities should spearhead the development of an integrated smart digital platform for meal assistance services. On the supply side, the platform should incorporate functionalities spanning the food supply chain, dynamic supply-demand dynamics, and last-mile delivery logistics, thereby facilitating the oversight and management of meal assistance outlets, enabling inter-regional resource reallocation, and augmenting overall utilization efficiency. On the demand side, the platform should integrate a suite of capabilities encompassing identity authentication, reservation, payment processing, subsidy administration, and health record management, to deliver unified identity verification, subsidy disbursement, and payment settlement, thereby attenuating cross-regional technical impediments. Second, a

"meal assistance card" initiative should be piloted and progressively rolled out. The cross-regional electronic payment and settlement infrastructure should be refined to ensure the expeditious circulation and interoperability of funds and associated datasets across jurisdictions. Adopting an incremental approach—commencing with targeted pilots and expanding to broader coverage—a number of central urban districts should be selected initially to launch universal pilots of the "meal assistance card," accruing experience and undertaking iterative refinements, with the ultimate objective of progressively realizing the principle that "benefits accompany the individual." Third, age-appropriate software development should be prioritized. With due regard for the physical capacities and customary usage patterns of older adults, user-centric design features—including enlarged typography and voice narration—should be incorporated, complemented by functionalities such as information inquiry, geolocation, emergency assistance, and service evaluation. These measures will streamline online reservation, payment, and customized meal ordering for older adults, rendering the cross-administrative sharing of high-quality meal assistance services technologically feasible. Concurrently, in recognition of the pervasive digital divide confronting the elderly demographic, peer mutual assistance among older adults should be fostered, and intergenerational "digital feedback"—whereby younger family members impart digital literacy—should be actively promoted. Public awareness campaigns and training programs should be conducted across society to popularize digital device and application proficiency, ensuring that older adults are not excluded from the benefits of smart public services.

4.4 Precise Demand Identification and Unblocked Feedback Channels

Demand identification and feedback function as the lubricant for the efficacious operation of the bidirectional embedding mechanism, the linchpin of which resides in accuracy and timeliness. First, big data analytics should be deployed to conduct periodic assessments of elderly mobility patterns and evolving demand trajectories for meal assistance services, and to establish comprehensive health records for older adults. This will enable precise demand identification and granular management, drive continuous optimization of elderly meal assistance services, surmount the constraints imposed by the "household registration–territorial jurisdiction" paradigm, and effect a paradigm shift from "physical proximity" to "service integration." Second, a diversified demand feedback mechanism should be instituted. Suggestion receptacles should be positioned in prominent locations at meal assistance venues, senior activity centers, and neighborhood committee offices, and staff should be periodically deployed to conduct home visit outreach, with particular solicitude extended to vulnerable elderly cohorts—including the advanced-age elderly, solitary older persons, and those with functional impairments. In the digital domain, feedback modalities should be embedded within WeChat communities, telephone hotlines, and dedicated meal assistance applications to systematically capture the expressed needs of older adults and their family members regarding meal assistance services. Third-party evaluations of elderly service satisfaction should likewise be commissioned and incorporated into the performance appraisal metrics of meal assistance venues and their respective jurisdictional authorities, thereby amplifying the responsiveness of the service supply side to elderly feedback, and ultimately realizing the vision that "services follow the elderly, resources revolve around the elderly." Third, the demand processing infrastructure should be fortified. Needs gathered through disparate channels should be taxonomically categorized—by meal offerings, service quality, delivery logistics, and individualized requirements—and routed to the appropriate operational units, including meal preparation facilities and delivery stations, for resolution. Processing outcomes should be periodically disseminated via community notice boards, meal assistance outlets, and official WeChat accounts. Older adults who submit actionable suggestions should be acknowledged with incentives such as discount coupons or modest tokens of appreciation, engendering a genuine sense that their voices are heard and valued, and thereby incentivizing further participatory engagement.

5 Conclusion and Discussion

Anchored in the theoretical foundations of collaborative governance, cross-boundary governance, and grassroots administrative district–community studies, this paper articulates a bidirectional embedding mechanism for administrative district governance and community life circles, thereby furnishing a conceptual framework for advancing the cross-administrative sharing of elderly meal assistance services.

Centering on the inherent tension between the institutional rigidity of administrative district governance and the demand elasticity of community life circles, and operating through the dual vectors of structural embedding and functional embedding, this mechanism impels state power to iteratively refine institutions and policies through sustained engagement with grassroots communities. In reciprocal fashion, community life circles exert a formative influence on administrative district governance through the articulation of needs and the mobilization of social participation. The concrete practical pathways encompass the strengthening of top-level design, the promotion of multi-stakeholder synergy, technology-enabled smart governance, and precision-oriented demand identification. As

population aging in China intensifies and the spatial mobility of older adults continues to rise, the cross-administrative sharing of elderly meal assistance services emerges as an inexorable trend. The task of expanding elderly meal assistance services from a model of "localized service provision" to one of "regional sharing" necessitates the dismantling of administrative barriers, the rationalization of resource allocation, and the elevation of service quality. It is anticipated that the bidirectional embedding mechanism advanced in this paper will furnish constructive insights for related theoretical inquiry and practical exploration, and will contribute to propelling elderly meal assistance services in China toward a new phase of high-quality development.

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About the author: Xiong Jing (1980–), male, is an Associate Researcher at the School of International and Public

Affairs, Shanghai Jiao Tong University, and Assistant to the Dean of the China Institute of Urban Governance. His research interests encompass spatial governance and administrative district systems in megacities and super-large cities.